MICONAZOLE

Oral gel raises INR

Infant choking risk

CHEMICAL NAME 1-[2-(2,4-dichlorophenyl)-2-[(2,4-dichlorophenyl)methoxy]ethyl]-1H-imidazole (nitrate)

CLASS Imidazole antifungal (topical, oral gel)

TRADE NAMES Daktarin · Monistat · Micatin · Daktacort (with hydrocortisone)

THERAPEUTIC USES / INDICATIONS

- Oral candidiasis (oral gel) in adults and infants >4 months.
- Skin fungal infections: tinea, candidal intertrigo, nappy rash.
- Vaginal candidiasis.
- Fungal nail infections (lacquer, limited).

ADMINISTRATION

- **Oral gel:** apply to affected areas after meals; hold in mouth before swallowing; remove dentures.
- Infants: apply small amounts to the front of the mouth — not to the back of the throat (choking).
- **Cream/powder:** apply twice daily to clean, dry skin.

DOSAGES

- **Oral gel (adult/child >2 yr):** 2.5 mL four times daily after food.
- **Infants 4–24 months:** 1.25 mL four times daily (divided into small portions).
- Continue for 7 days after symptoms resolve.
- **Skin:** twice daily for 2–6 weeks.

PHARMACOKINETICS

- Oral gel partly absorbed systemically (~25%); topical minimal absorption.
- Metabolised in liver (strong CYP2C9 and 3A4 inhibitor when absorbed).

PHARMACODYNAMICS

- Inhibits fungal **ergosterol synthesis** → membrane damage.
- Active against *Candida*, dermatophytes; some Gram-positive bacteria.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, unpleasant taste (gel).
- Local irritation, burning (skin).

ADVERSE EFFECTS (SERIOUS)

- Choking in infants (gel).
- Marked rise in INR / bleeding with warfarin (even oral gel and vaginal forms).
- Allergic reactions.

CONTRAINDICATIONS & CAUTIONS

- Infants <4 months (oral gel) or with impaired swallowing.
- Use with warfarin (oral gel; avoid or monitor closely); simvastatin, ergot alkaloids, pimozide.
- Liver dysfunction (oral gel).

KEY INTERACTIONS

- **Warfarin:** serious ↑ INR.
- **Sulfonylureas:** hypoglycaemia; **phenytoin:** ↑ levels.
- **Statins:** ↑ myopathy.

NURSING MANAGEMENT

- Check for warfarin use before giving oral gel; monitor INR if needed.
- Teach parents safe gel application for infants.
- Inspect mouth for white plaques; check dentures.
- Treat breastfeeding mother's nipples if infant has thrush.

PATIENT EDUCATION

- Use the gel after feeds/meals, in small amounts.
- Clean dentures daily and soak overnight.
- Continue a week after spots disappear.
- Tell your doctor if you take warfarin.

KEY POINTS

- Imidazole — oral gel for oral thrush.
- Infants: small amounts to front of mouth (choking).
- Interacts strongly with warfarin.
- Not under 4 months.